Rana Mall Singh

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OBJECTIVE

Full Stack Developer with 2+ years of experience working with React.js, Node.js, and MongoDB. Good at making web apps that work well and are fast. Proven ability to improve the user experience and speed up the performance of applications. Passionate about coming up with new ideas and eager to work with others in a fast-paced environment.

EDUCATION

Bachelor of Computer Science & Engineering, Chandigarh Group of Colleges 2022 - 2026

TECHNICAL SKILLS

Programming Languages: JavaScript, C++, PHP, Python, Solidity
Frontend: HTML, CSS, JavaScript, React, Next.js, Redux, Bootstrap, Tailwind CSS, jQuery, MUI.
Backend & APIs: Node.js, Express.js, REST API, GraphQL, microservices.
DevOps & Tools: Docker, Webpack, Parcel, Git, GitHub
Databases: MongoDB, MySQL
Real-Time & Streaming: WebSocket
Blockchain: Smart Contracts, Solidity
Deployment: Heroku, Netlify, Vercel
Design & Prototyping: Figma
Operating Systems: Linux

WORK EXPERIENCE

SR4DX August 2024 – Present

Software Engineer – Frontend

Mizukara (Client Project – Water Treatment & Clean Energy, Japan)

1. Inspection Master System

- Architected and built a dynamic form and workflow engine using **Next.js (with SSR/ISR)**, **React Hook Form**, and **Redux Toolkit**, enabling admins to create, edit, schedule, and assign inspection forms across teams.
- Integrated real-time form status tracking and audit trails using **Socket.IO** and custom event listeners, allowing controllers to monitor and review filled forms instantly.
- Included a scheduler module with support for bundling multiple forms, assigning deadlines, and merging inspection types.

2. Notification Hub

- Implemented a configurable multi-channel notification system, enabling communication through **WhatsApp (Cloud API)**, **Twilio SMS**, **LINE SDK**, **Slack Webhooks**, **Chatwork**, and **Email (SMTP)**.
- Built a flexible routing layer with conditional logic to trigger alerts for form deadlines, API hits, and system events.
- Made notification preferences and delivery modes configurable per user and per organization, ensuring global usability.

3. API Gateway

- Developed a full-fledged **API Explorer** that let users define, test, and use custom APIs to fetch real-time organizational data (e.g., machine performance, reports, inspection logs).
- Created a **Platform API panel** with a Swagger-inspired UI and JWT-secured access, where users could monitor, document, and share their APIs.
- Designed a **Business API interface** where clients could request new APIs from the backend team through a governed workflow.

4. Report Builder

- Built a **low-code report builder** with JSON-configurable layouts using **Syncfusion**, **Chart.js**, and **pdfmake**, allowing users to generate visually rich reports based on selected data sources.
- Added support for advanced features like dynamic data grouping, aggregation logic, formatting preferences, and real-time previews.
- Enabled **automated scheduling** of reports (daily, weekly, custom CRON) and multi-format export (PDF, DOCX), with delivery via email or internal platform.

5. Dynamic Dashboards

- Designed a drag-and-drop dashboard builder using **Grid Layout** and **react-query**, letting users create fully personalized dashboards with real-time charts, KPIs, and form/API widgets.
- Integrated role-based visibility and live data streams with lazy loading, ensuring performance and personalization across dashboards.
- Supported embedding reports, API metrics, inspection statuses, and any visual module across user profiles.

6. Template Builder

- Developed a visual **template-building module** to define performance tracking formats for plant machinery, allowing admins to map metrics, configure inputs, and reuse templates across systems.
- Enabled dynamic field binding to real-time API data, with formula logic for computing machinery health or treatment values.
- Integrated this builder with the reporting and dashboard modules for seamless performance monitoring.

7. Log Insights

- Implemented an **activity monitoring system** that tracked every interaction across the platform from form submissions to API edits using contextual metadata and timestamped entries.
- Enabled organizational-level health tracking, improving transparency, traceability, and compliance for client stakeholders.

EV Charging Connectivity App (IR App)

1. Admin Panel:

- Built a live **charging station dashboard** using **Leaflet.js** and **OpenStreetMap**, supporting real-time updates, device clustering, and geolocation-based filtering.
- Added full user & billing management along with a **role-based access control system** to restrict features at a granular level.
- Integrated real-time usage logs, energy consumption data, and reports into the admin view.

2. Consumer App:

- Developed a **QR-based charging flow**, allowing users to scan, choose payment (via **Stripe** or **Razorpay**), and initiate the charging session.
- Built session management logic with **local storage caching**, automated disconnection triggers, and lifecycle-aware UI.
- Embedded games and ad modules during the charging process to enhance user engagement.

CYBERFLOW April 2021 – August 2021

Frontend Engineer

Employee Tracker Web Application:

- Developed and deployed a scalable Employee Tracker with Electron.js, React, Redux, and Socket.io, enabling real-time activity monitoring and instant updates.

Admin Panel Integration:

- Designed and implemented a responsive Admin Panel within the Employee Tracker, utilizing CSS, React, React-Redux, Syndash Library, Syncfusion, Material-UI, Bootstrap, and Chart.js. Tailored controls for diverse user roles, enhancing user experience and ensuring smooth interaction with the application.

PROJECTS / OPEN SOURCE

Vote Chain: A Blockchain-Powered Voting System

- Innovative implementation of a secure and transparent voting system using blockchain technology. This project, crafted during an MLH hackathon and securing a spot in the top 10, aims to revolutionize election processes by ensuring trust, integrity, and accessibility. Utilized technologies include React, Volta (Testnet), Bootstrap, Solidity, and MetaMask.
- GitHub :https://github.com/Ranamalsingh12/Votechain

Crypto Verse: Comprehensive Cryptocurrency Information

- A React.js-based website equipped with Redux-Toolkit, Millyfy, Chart.JS, HTML-react-parser, Moment, Ant Design, Axios, and UUID. Offering pages dedicated to cryptocurrency news, the top 100 cryptocurrencies, historical data, and performance graphs, CryptoVerse provides a holistic resource for crypto enthusiasts and investors alike.
- Github :https://github.com/Ranamalsingh12/CryptoVerse

Open-Source Contributor – React.js (by Meta)

Contributed to the official React.js codebase by **improving and refining the useEffect documentation**, focusing on **dependency array behavior, execution timing**, and **edge case explanations**. The update helped improve clarity for both beginners and experienced developers referencing the official docs. Actively engaged in the React community during the review process to ensure accuracy and consistency with React's core principles.

ARTICLE / BLOGS

Explore my latest article, titled 'The Execution Context of JS,' available on Hashnode.

Link to Aricle : https://ranamsblogs.hashnode.dev